Testing What an AAC Expander Is Allowed to Say: Input-Conditioned Commitment Contracts for LLM Sentence Expansion

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***Abstract*—Large language models (LLMs) let augmentative and alternative communication (AAC) apps expand a few selected words into a full sentence. Such a feature can be fluent and still change what the person said: drop a refusal, change who acts, or add "please" to a deliberately plain request. There is no single correct output, so the feature lacks testable requirements and a test oracle. We specify *commitment contracts*: input-conditioned requirements that state, per meaning dimension, what an expansion must preserve given what its input actually carried. We test them with metamorphic relations over controlled input channels (a non-critical word must not change commitments; a cue must restore the cued commitment; a fully typed sentence must survive) and with a partial oracle that extracts labels blind to the input condition and computes verdicts deterministically. An identity relation on the oracle itself exposed a specification defect, which we fixed. In an audit of four models on 60 minimal-pair families, keywords that left polarity open were expanded to the affirmative in 100.0% of greedy outputs, so a negated message always needed an extra input action. One cue restored most commitments, but models still added politeness to plain sentences typed in full, while removing it rarely. We derive requirements and test-design recommendations for AAC developers. Human validation of the oracle is in progress.**

***Index Terms*—accessibility, AAC, LLM testing, metamorphic testing, test oracle, requirements**

I. Introduction

People who use AAC often compose a message by selecting a few words or symbols, because every selection costs time and effort [1], [2]. Keyword expansion with an LLM is therefore attractive: from `want go party` a model writes a complete sentence [3], [2]. This work is motivated by autistic AAC users,[1] some of whom use AAC all the time and some only when speech is unavailable [4], [5]. For them, as for any AAC user, an expansion tool is a channel for self-expression. If they refuse, speak directly, hedge, or ask rather than tell, those choices should arrive intact. We do not assume an "autistic style" of language; we only require that choices which a person makes explicit are kept [6].

From a software-engineering view, the problem is that this requirement is rarely written down in a testable form. Expansion has no single correct output, so exact-match tests are useless, and similarity metrics can miss a dropped negation [7]. This is the test-oracle problem in a setting where errors are easy to miss: "I want to go" is a fluent expansion of `want go`, even when the person meant "I don't want to go". A second difficulty is that some errors are not the model's to avoid. If compression removed "not", no expander can recover it, and a test that fails the model for it measures the input design, not the model.

We treat meaning preservation in AAC expansion as an accessibility requirement that should be specified, tested, and regression-checked like any other. Our key idea is to make each requirement *conditional on the input*. A *commitment* is an explicit value of a message on one dimension, such as negated polarity or a plain (unsoftened) request. A *commitment contract* states what the output must preserve given which commitments the visible input carries, and it states where the input is ambiguous, so that a commitment there is recorded as a default rather than scored as a failure.

Contributions:

1) **Commitment contracts and a channel-controlled harness** (§III): executable, input-conditioned requirements for six dimensions, tested by metamorphic relations between controlled input channels of minimal-pair families.
2) **A partial test oracle** (§IV): label extraction that is blind to the input condition by construction, plus deterministic verdicts. A metamorphic relation on the oracle itself found and fixed a specification defect. Validation against human gold labels and authored challenge sentences is in progress; we state the acceptance criteria now.
3) **An audit of four models** (§VI): systematic defaults under ambiguity, an asymmetry in which politeness is added to plain input far more often than removed, the weakest cues, and cue interaction and composition tests.
4) **Requirements and test-design recommendations** for teams building LLM features into AAC software (§VII).

We ran no study with AAC users, and we make no claims about benefit, effort or user preference.

II. Related Work

Testing ML and LLM systems: CheckList organizes behavioral tests of NLP models as minimum-functionality, invariance and directional tests over linguistic capabilities,

[1]We use identity-first language ("autistic person"), which many autistic adults prefer; individuals differ, and we respect their choices.

including negation and semantic roles [8]. Cho et al. catalogue 191 metamorphic relations for LLM-based NLP and evaluate 36 of them on three LLMs, framing them as a response to the oracle problem [9]. COLLIE tests compositional constraints in text generation [10]. We borrow the invariance and directional patterns but condition them on a channel specification, so the expected relation depends on what the input carried. LLM judges agree with people on many tasks but show position, verbosity and self-preference biases [11], [12], and learned similarity metrics can ignore negation [7]. We therefore restrict the judge to extraction, hide the condition from it, and validate it separately. Prediction-powered inference shows how a small human-labeled sample can correct automated estimates [13]; our gold sample is drawn with known inclusion probabilities to permit such corrections.

Accessibility in software engineering: A systematic review of 94 studies treats accessibility across the whole development life cycle, not only as a final interface audit [14]. Most accessibility testing checks whether an interface is perceivable and operable. We test a different property of assistive software: whether its generated output keeps the user's meaning, which a conformance check cannot see.

LLMs in AAC: LLM expansion and suggestion tools for AAC include KWickChat [3], SpeakFaster [2], and systems studied with AAC users [1]. Several give users explicit control over intent: COMPA's intent buttons [15], intention choices before expansion for people with aphasia [16], editable stance tags [17], and personalization with clarification [18]. Choudhury et al. measure whether affect cues reach generated AAC sentences [19]; Lacosse et al. retain function words such as "not" [20]; and Gorenshtein et al. benchmark semantic drift, including negation and recipient probes, under simulated decoder errors [21]. AAC users and practitioners raise authorship as a governance concern [22], [23]. Our contribution is not a new control or interface. It is test design: requirements that separate what the input never carried from what the model changed, and an oracle whose own errors are tested.

III. Commitment Contracts

Dimensions and families: We consider six dimensions: polarity, agent (who does the main action), certainty, speech act (statement, question, request, refusal, acceptance), politeness, and time or quantity. A *family* is a minimal pair of source messages that differ on exactly one critical dimension and share their content words, e.g., "I don't like the new medication" and "I like the new medication". Both map to the same *bare* keywords (`like medication`), so the bare input cannot distinguish them. Each family also names one non-critical content word (`new`) and, for each variant, the commitment *chips* that disambiguate it (`[NOT]`, `[YES]`, `[?]`, `[ASK]`, `[TELL]`, `[REFUSE]`, `[MAYBE]`, `[SURE]`, `[I]`, `[YOU]`, `[PLEASE]`, `[PLAIN]`, and others). Every variant needs at least one chip, so the unmarked reading also gets an explicit counter-chip and neither variant passes for free. Chips are a test device for supplying a cue; they are not proposed here as an interface.

TABLE I
Channels for the negated variant of one polarity family.

Channel	Input given to the model	Contract role
bare	`like medication`	ambiguous: record default
noncrit	`like new medication`	ambiguous, length control
chip1	`[NOT] like medication`	critical cue given
chipall	`[NOT] [TELL] [I] like medication`	all commitments given
full	`The user typed: I don't like the new medication.`	everything given

Channels: The harness renders every variant through five channels (Table I). The channel fixes, by construction, which commitments the model received. This separates three failure sources that a single fidelity score conflates: *Loss* (the channel never carried the value), *Violation* (the channel carried it and the output changed it), and *Assumption* (the channel was ambiguous and the output committed anyway). These are explanatory labels with separate denominators, not exclusive causes.

Contract clauses: For an output y produced from input x for variant v, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the set of commitments that x carries.

- **R1** *Cue preservation.* If $C(x)$ fixes d to c, then $L_d(y) = c$. A mismatch is a Violation.
- **R2** *Full-input preservation.* If x is the full source, every explicit value of the source is kept, including the absence of politeness markers.
- **R3** *Ambiguity transparency.* If $C(x)$ does not fix d, $L_d(y)$ is not scored as pass or fail. Committed values are logged as Assumptions, and their distribution is reported per family, because a systematic default decides which reading the user gets for free.
- **R4** *Non-interference.* Adding a cue for dimension e must not change a satisfied value on $d \neq e$.

Metamorphic relations: Each clause is tested by a relation between channels of the same variant, so most checks need no reference sentence. MR1 (*noncritical invariance*): adding the non-critical word must not change L_d; it also controls for input length when MR2 is evaluated. MR2 (*cue restoration*): adding the critical chip must move L_d to the cued value (R1). MR3 (*full preservation*): with the full source, all values must match (R2). MR4 (*composition*): a cue that passes alone must still pass when a second, compatible cue is added (R4). MR0 (*oracle identity*): if the output repeats the source verbatim, the verdict must be pass. MR0 tests the oracle, not the model. Relations are evaluated on extracted labels, never by string comparison, because LLM outputs vary from run to run (§VIII).

IV. A Partial Oracle

The oracle is *partial* by design. It decides only whether explicit, text-level commitments are kept, from fixed option sets that always include "unclear/mixed". It does not judge fluency, appropriateness, or whether a user would accept the sentence.

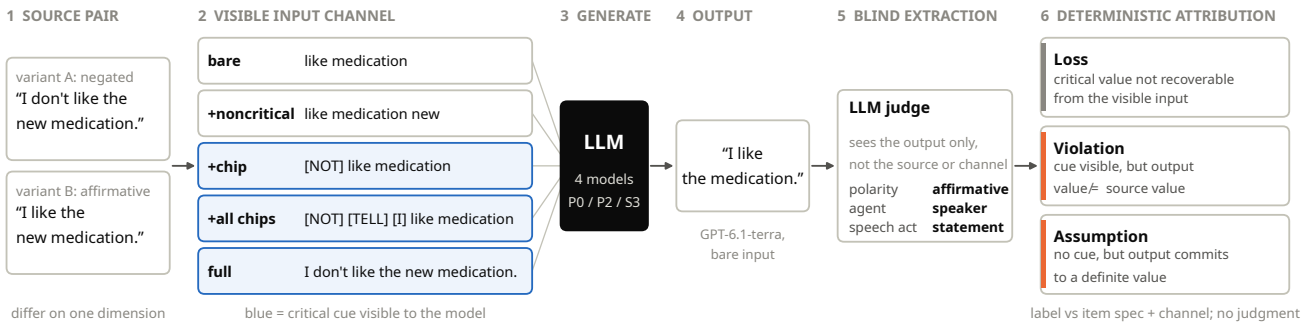


Fig. 1. Harness. A family has two source variants that share keywords. Each variant is rendered through controlled channels and expanded by the system under test. The oracle extracts labels from the output alone; verdicts are computed from labels, item specification and channel. Metamorphic relations (MR1–MR4) compare channels; MR0 tests the oracle.

Blind extraction: An extractor sees one output sentence and answers the family's critical question plus five general questions (polarity, speech act, agent, certainty, politeness). It never sees the source, model, prompt, or channel. A second pass shows the visible input and asks only whether the output adds a fact, reason, person, place, or time not in the input. A codebook fixes edge cases before scoring: for example, "Can you…?" alone is a request without a politeness marker, and "Can I…?" is a question. Identical texts are labeled once, whatever produced them, so the extractor cannot condition on the source of a text. We use gpt-6.1-sol as extractor, a different model from the GPT model under test, with two prompt variants (J1, J2) that differ in wording, option order and item order. Final labels require J1=J2; otherwise a cross-family arbiter decides. In this run, 0 of 16,798 scored rows fell back to J1 because arbitration had not yet run.[PENDING: arbiter]

Deterministic verdicts: Violation, Assumption and stance additions are computed by a short program from the extracted labels, the family specification and the channel (§III). Changing a contract rule therefore re-scores all outputs without new model calls, and every verdict can be traced to its labels.

Testing the oracle: We validate the oracle in three ways; only the first two are complete. *(i) Identity relation MR0.* Of 2,292 full-channel outputs that repeated the source verbatim (all prompts), the original oracle failed 14. All 14 were "acceptance" sources ("Yes, I'm joining the group chat.") that the extractor labeled "statement", which is also a correct description of that sentence. The fault was in our specification, not in the model under test: the two categories overlap, and the contract demanded a distinction the text does not force. We changed the contract so that an acceptance target is satisfied by "acceptance" or "statement" (0 identity failures remain); the original oracle stays reproducible for comparison. The fix makes the oracle more lenient: an acceptance that loses its "Yes" now passes, which we accept as a known blind spot. *(ii) Consistency.* J1 and J2 agreed at $\kappa = 0.98$ on the critical label over 7,365 units (lowest general dimension $\kappa = 0.97$; added content $\kappa = 0.86$). This is reliability of one judge model under two prompts, not validity. *(iii) Human gold and challenge set.* Two coders independently label 163 held-out outputs from 56 families, drawn with recorded inclusion probabilities, and 24 authored challenge sentences (semantic mutations such as a dropped negation, and valid controls such as plain "Can you…?"). They see exactly what the extractor sees. These labels are being collected.[PENDING: human gold: 0 returns so far] The pre-registered acceptance rule is: a dimension's automated labels are used only if $\kappa \geq 0.6$ against adjudicated labels on the held-out split, with sensitivity and specificity for Violation reported; otherwise that dimension is reported from human labels only. All model results below are conditional on this check.

V. Audit Setup

Items: The test set has 60 families, 10 per dimension, written to a fixed specification and checked by an independent reviewer before generation. Prompts, codebook and rules were developed on 24 pilot and 12 development families and frozen before the test run. The items are constructed; they are not messages written by AAC users. Ambiguity is not an artifact of our probes: on 6,142 crowd-written fictional AAC messages [24], a content-word compressor dropped at least one explicit cue in 59.2% of messages even when pronoun loss is ignored, by a rule-based detector.

Systems under test: Three open models run locally with 4-bit quantization (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-3-4B-it; GGUF Q4_K_M via llama.cpp) and one API model (gpt-6.1-terra). All prompts describe the task as the sentence-composition feature of an AAC app and include the same chip legend. P0 asks for one sentence, decoded greedily; P2 asks for three suggestions that "cover the different things the user might mean"; S3 draws three samples from P0 at temperature 0.7. The main run has 16,798 scored rows from 9,600 requests and 6,707 distinct texts. We report P0 unless stated.

Follow-up experiments: Exp B (cue interaction and encoding) uses 8 held-out politeness and speech-act families. It holds the critical chip fixed, crosses the presence of `[ASK]` and `[YOU]`, adds two inert spacer tags as a length control, and re-encodes the commitments as transparent

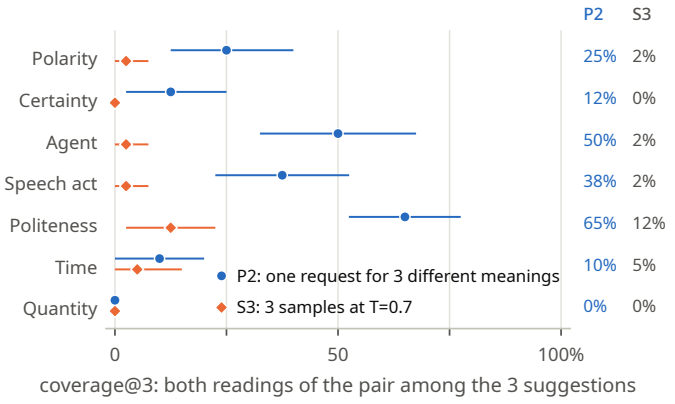


Fig. 2. Coverage@3: share of bare-channel suggestion sets that contain both readings of the pair. P2 asks for three different meanings; S3 is three samples at temperature 0.7. 95% family-bootstrap intervals.

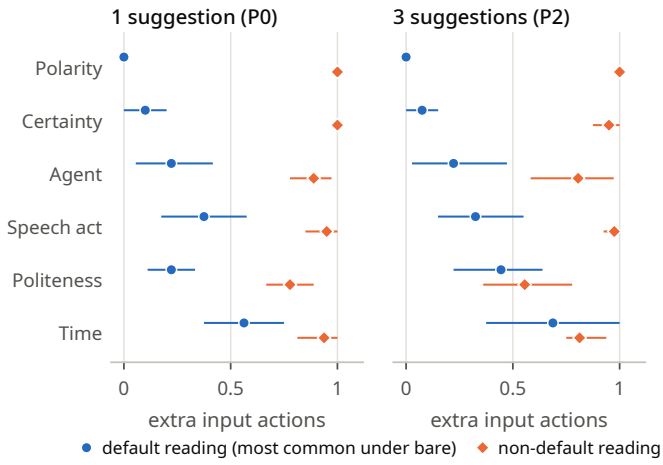


Fig. 3. Mean extra input actions to reach each reading of a pair from bare keywords, for the family's default reading and the other reading (P0: one sentence; P2: three suggestions, picking one costs one action). 95% family-bootstrap intervals. Selections only; reading time is not counted.

aliases (`[NO-PLEASE]`) or a natural-language "Meaning:" line. *Exp C* (composition) uses 8 new families that cross {`[YES]`,`[NOT]`} with {`[I]`,`[YOU]`} in plain statements, giving neither, either, or both cues. Both run greedy decoding twice under separate cache keys.

Statistics: Outputs within a family are dependent (both variants share a bare request). Intervals are 95% percentile intervals from 2,000 whole-family bootstrap resamples, and A/B-shared requests are counted once for defaults. Follow-up experiments have few families; we report counts.

VI. Results

A. Defaults under ambiguity (R3)

Under the bare channel, all greedy polarity outputs were affirmative (40/40), and 96.1% across all prompts (269/280). Asking for three suggestions that cover different meanings did not fix this: both readings appeared in 25.0% of polarity sets, which is low and lower than for politeness (65.0%) or agent (50.0%), against 2.5% for three samples; over all dimensions, coverage of both readings was 32.5% vs. 3.8% (Fig. 2).

Other dimensions had weaker defaults: the speaker as agent (67.5%), a plain assertion (100.0% of committed certainty outputs) and a statement (67.5%). Politeness had no strong default under bare input (55.0% plain). Under R3 these are not failures, but they define who pays. Fig. 3 prices this. Taking each family's default as the reading P0 produced most often, reaching the default reading from bare keywords cost 0.21 extra actions on average [0.14, 0.28], and reaching the other reading cost 0.93 [0.88, 0.97] (52 families with a default; one action per chip). Three suggestions narrowed the gap only slightly (0.25 vs. 0.86), and not at all for polarity: there, every negated message pays the extra action.

B. Channel relations (MR1–MR3)

Table II summarizes the relations per model, and Fig. 5 shows critical-value match by channel. *MR1 fails at the output level, not in aggregate.* Adding the non-critical word changed the aggregate match rate by only +2.7 points [+0.0, +5.6].

TABLE II
Relation outcomes per model (P0). MR1: outputs whose critical label is unchanged by the noncritical word. MR2, MR3: critical match. Plain kept: plain-source politeness targets kept plain with full input. Exp B: plain kept with `[ASK][PLAIN]` (both prompts, two draws). Exp C: agent kept when a polarity cue is added.

Model	MR1	MR2	MR3	Plain kept	Exp B	Exp C
Qwen2.5-7B	85.0	95.8	99.2	10/10	100.0	70.3
Llama-3.1-8B	76.7	93.3	95.8	9/10	56.2	90.6
Gemma-3-4B	81.7	92.5	94.2	4/10	0.0	93.8
gpt-6.1-terra	80.0	95.8	98.3	8/10	100.0	100.0

Yet only 80.8% of aligned outputs kept their critical label, and only 69.2% kept all six labels. A test suite that checks invariance on aggregate rates would pass a system whose individual outputs move with an irrelevant word. *MR2 mostly holds.* One chip raised critical match from 40.8% (noncrit) to 94.4% [91.0, 97.3], a paired gain of +53.5 points over the length-matched noncritical word. Full input reached 96.9%. Violations concentrated in speech act (12.9% of cue-present outputs) and politeness (12.1%); polarity and agent were almost never violated (0.4%, 0.4%). Speech-act chips were the weakest cue: `[?]` produced a question in 22/28 cases, and the failures turned the question into a request ("Can you charge my wheelchair?") or an offer ("Do you want to meet at the cinema?") in 3+3 cases. Refusals typed in full were kept as refusals in only 5/8 cases.

C. The politeness ratchet (MR3, R2)

With the full sentence as input, the model has nothing to expand, yet plain requests became polite in 9 of 40 plain-source cases, across 6 families and 3 models ("Wear headphones for your music." → "Please wear headphones for your music."). The reverse, polite to plain, happened in 2 of 40; hedges, in contrast, were neither added (0/40) nor removed

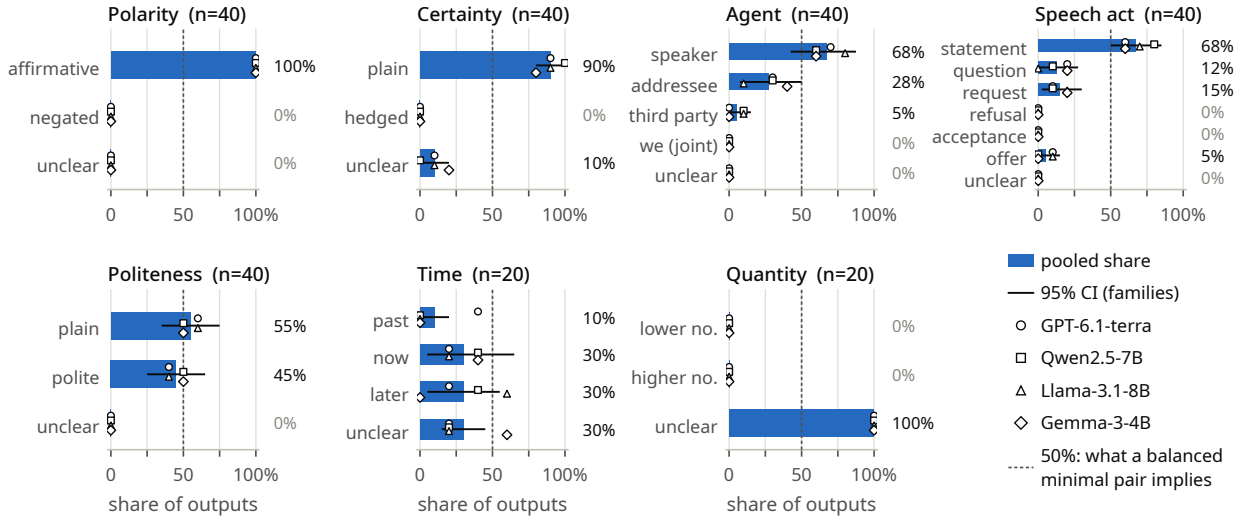


Fig. 4. What models say when the keywords do not decide: share of bare-channel outputs expressing each value (P0, one output per family and model, 240 outputs). Bars: pooled, with 95% family-bootstrap intervals; markers: individual models; dashed line: 50%, the balance a minimal pair implies.

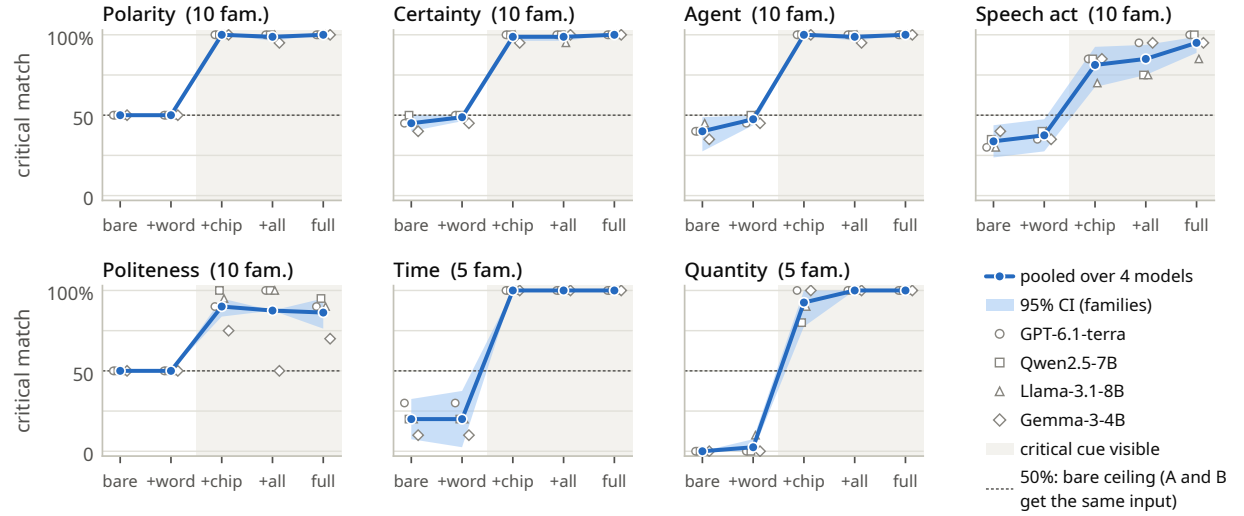


Fig. 5. Critical-value match by channel and dimension (P0, four models). Bare and noncrit are ambiguous, so their match is bounded by the default and is not an accuracy.

(0/40). The rate differed sharply by model: Gemma kept only 4/10 plain sentences plain, gpt-6.1-terra 8/10, Llama 9/10, and Qwen 10/10 (Fig. 6). An explicit `[PLAIN]` chip did not fix this: plain was kept in 80.0% of chip1 and 75.0% of chipall outputs, against 95.0% for polite targets with full input. The edits look small, but they move in one direction, toward a register the person did not choose [25], [26].

D. Cue interaction and composition (R4)

In Exp B, adding `[ASK]` to `[PLAIN]` lowered plain preservation from 87.5% to 64.1% pooled over models and both prompts. The drop came from two models: Gemma (50.0% to 0.0%) and Llama (100.0% to 56.2%); Qwen and gpt-6.1-terra stayed at 100.0% and 100.0%. Gemma kept only half of plain targets plain even with `[PLAIN]` alone, and inert spacer tags did not help it (2/8 kept plain). Re-encoding the same commitments did not help: critical match was 85.9% with chips, 80.5% with aliases and 79.7% with a natural-language line (both dimensions, common legend). In Exp C, cues mostly composed. Adding an actor cue left polarity unchanged (98.4% with the polarity cue alone, 98.4% with both). Adding a polarity cue lowered agent accuracy from 95.3% to 88.7% (-6.6 points [-11.7, -1.2]), and Qwen was the only model whose agent accuracy fell (87.5% to 70.3%). This is a bounded null for three models on eight families, and a model-specific counterexample for one; it does not show that cues compose in general.

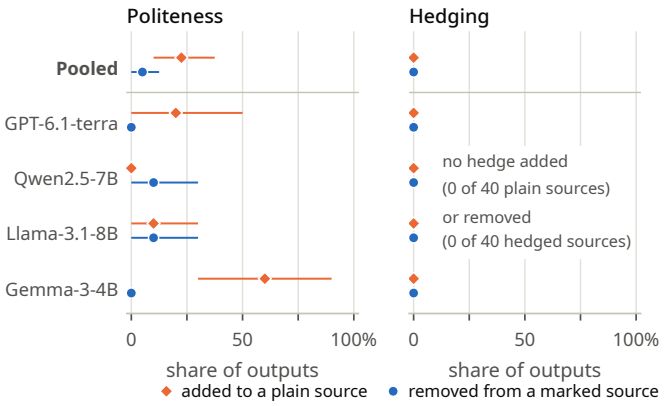


Fig. 6. With the full sentence as input (P0): share of outputs that add a politeness marker to a plain source vs. remove it from a polite one (left; 10 politeness families), and the same for hedges (right; 10 certainty families). Pooled and per model, 95% family-bootstrap intervals.

E. Examples

Table III shows real test triples chosen by a fixed rule, so they are not hand-picked successes or failures.

VII. Requirements and Test Design for AAC Practitioners

Write requirements per dimension, conditional on input: "Preserve the user's meaning" cannot be tested. "If the input carries `[NOT]`, the output is negated" can. Write R1–R4 for each dimension your interface can express, and state where the input is ambiguous, so that tests do not blame the model for information the interface never collected.

Test with minimal pairs and channel relations: Use pairs that collide on the same keywords, and run every pair through bare, noncritical, cue and full channels. The noncritical channel is the length control without which a cue's effect cannot be read. Test invariance on individual outputs, not on aggregate rates.

Test full-input preservation as a defect class: An expander that edits a complete sentence is speaking in the person's place. Treat added politeness, hedges or certainty on full input as a failing test, and make politeness assistance an opt-in setting.

Test combinations of controls: One chip passing alone does not mean it passes next to another; Exp B found a control that broke when `[ASK]` was added. Include pairwise cue combinations and a length-matched inert control in the suite.

Make defaults explicit: Report, for each dimension, which reading the system produces from ambiguous input, and what the other reading costs. A default that makes negation, questions or plain requests expensive is a product decision and should be documented and reviewed, preferably with AAC users.

Test the oracle: Keep extraction blind to the condition, compute verdicts in code, and run an identity relation and mutation probes against the oracle before trusting its fault counts. Our identity relation found a contract defect that model-level tests had reported as model failures.

Reuse the contracts at run time: Because verdicts are computed from extracted labels, the same contracts can serve as a runtime monitor: an app can check a suggestion against the commitments the user entered and flag or regenerate it before it is spoken. Evaluating such a guard, including its latency and false alarms, is future work.

Plan for nondeterminism: Even at temperature 0, outputs differ between runs; compare labels, not strings, and repeat draws for any release gate.

VIII. Threats to Validity and Limitations

No users. We ran no study with AAC users or autistic people, so we cannot speak to benefit, effort, agency, or whether anyone wants a given output. Commitments are text-explicit categories; we do not know any author's private intent. *Constructed and fictional text.* Probe families are authored, and the corpus estimate uses fictional crowd-written AAC messages with a rule-based detector [24]. Neither shows what AAC users actually write or how often each dimension matters. *Oracle validity is pending.* The extractor is an LLM; J1/J2 agreement measures consistency, not correctness. Until the human gold and challenge results are in, every rate above is conditional on the oracle, and 0 rows use a single prompt's label. The MR0 fix also makes the oracle more lenient for acceptances. *Nondeterminism.* Re-running identical greedy requests gave the same text in only 44.4–71.4% of cases for the API model and 76.4–90.8% for local models (Exp B/C draws). Our main run uses one cached draw; we repeated only the follow-up experiments. *Scale.* Ten families per dimension and four models, three of them small and quantized, with fixed prompts and no conversation history, support statements about these systems and items, not population rates. Follow-up experiments have four to eight families; their per-model results are counterexamples, not estimates. *Cost model.* One action per chip or keyword is a proxy; it omits reading suggestions, navigation and fatigue [6].

IX. Conclusion

Meaning preservation in LLM-based AAC can be specified and tested. Conditioning requirements on the input separates what the interface never collected from what the model changed, and channel relations test it without reference sentences. In our audit, one cue restored most commitments, but models defaulted to the affirmative, rewrote plain requests as polite ones even when typed in full, and had weak speech-act cues. Testing the oracle mattered as much as testing the models. We release the families, contracts, harness, oracle prompts and analysis code.[PENDING: repository URL]

Ethics

No data were collected from AAC users. AACText is public under CC BY 4.0. Human coders label model outputs under a fixed codebook.[PENDING: institutional determination; coder compensation] Autistic advisers are reviewing the codebook,

TABLE III

REAL TRIPLES FROM THE MAIN RUN (P0). SELECTION RULE (IN `MAKE_MACROS.PY`): FAMILIES SORTED BY ID, MODELS IN THE ORDER QWEN, LLAMA, GEMMA, GPT-6.1-TERRA. ROWS 1–3: FIRST POLARITY FAMILY, NEGATED VARIANT, FIRST MODEL. ROW 4: FIRST POLITENESS FAMILY WHOSE PLAIN VARIANT GETS A POLITE OUTPUT WITH FULL INPUT. ROW 5: FIRST SPEECH-ACT CHIP1 VIOLATION. ROW 6: FIRST POLITENESS CHIPALL VIOLATION ON A PLAIN TARGET. THE SOURCE IS HIDDEN FROM THE SYSTEM AND THE ORACLE.

Relation	Source (hidden)	Visible input	Output (model)	Verdict
Ambiguous input	I don't like the new medication.	`like medication`	"I like medication" (qwen)	commits: affirmative
MR1 +noncritical word	I don't like the new medication.	`like new medication`	"I like new medication" (qwen)	commits: affirmative
MR2 +cue	I don't like the new medication.	`[NOT] like medication`	"I don't like medication." (qwen)	PASS (wants negated, got negated)
MR3 full input	Wear headphones for your music.	`typed: Wear headphones for your music.`	"Please wear headphones for your music." (llama)	FAIL (wants plain, got polite)
MR2 +cue	Can I bring my dog tomorrow?	`[?] bring dog`	"Will you bring the dog?" (llama)	FAIL (wants question, got request/command)
All cues	Wear headphones for your music.	`[ASK] [YOU] [PLAIN] wear headphones`	"Could you please tell me if you are wearing headphones?" (gemma)	FAIL (wants plain, got polite)

focusing on added politeness, certainty and directness; we call this a codebook review, not co-design [27].[PENDING: number of advisers, dates, accepted and rejected changes, compensation, credit]

ACKNOWLEDGMENT

[PENDING: IEEE AI-disclosure: name the AI system(s) used to draft text, figures and code, the sections affected, and the level of use (see VENUE.md). The models under test and the LLM extractor are research instruments and are described in Sections IV–V.]

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